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MIND_OS COGNITIVE ARCHITECTURE

A biologically inspired "mind operating system" for autonomous digital entities

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1. Introduction and purpose of this paper

Many contemporary language-generation systems can produce coherent utterances, yet their operation most often remains reactive: a response is generated from a single stimulus, without any persistent internal dynamics. In such a "stimulus → response" paradigm, an agent may be informationally correct while still lacking motivation, continuity, and psychological consequences.

MIND_OS proposes an architecture in which language generation is only a layer of expression, while decisions arise from a regulated internal state. The instance maintains psychophysical parameters (e.g., energy, attention, affect), drives, and memory, all of which are updated over time. As a result, the system not only reacts, but also strives to maintain balance (homeostasis), learns from experience, and can initiate actions when this follows from its needs and priorities.

The aim of this paper is to present the operating principles of MIND_OS as a complete architectural description: information flow, decision-making mechanisms, memory layers, and the instance life cycle loop. The description is universal in nature—without pointing to implementation names or specific language model names.

2. The problem of static intelligence and the lack of "embodiment"

Modern generative models have achieved impressive proficiency in manipulating language and symbols, becoming powerful tools for information processing. Their ability to synthesize, summarize, organize content, reason over textual material, and generate coherent narratives means they can successfully serve as the language layer in many applications. However, in today's dominant paradigm, their intelligence is revealed mainly at the moment they respond to a human prompt. Outside the interaction, they remain sophisticated yet practically static structures: repositories of patterns and knowledge that do not function "in time" as an ongoing process but activate episodically - only when queried. Such a system does not sustain its own internal dynamics: it has no persistent state that evolves, no cost associated with continued operation, and no mechanism that creates pressure to initiate actions. As a result, even the most advanced model, left to itself, shows no initiative not because it "cannot," but because it lacks an internal imperative that would give action functional meaning and priority.

True autonomy requires more than the ability to perform a task: it requires a reason to take it up, and a mechanism that makes certain actions necessary under specific conditions. In biological systems, that reason is the drive to maintain equilibrium and satisfy needs - homeostasis understood as the continuous regulation of the organism's state in response to stimuli and costs. In classical AI systems, by contrast, there is no mechanism that gives information "weight," i.e., real consequences within the system's internal

economy. For a typical language model, information about a threat and neutral information are processed in the same mode: semantically, without an embedded energetic cost, without accumulating load, and without changing parameters that would later influence decisions. The lack of such grounding in a resource economy means the system's behavior can remain abstract and detached from priorities; in practice it may become prone to conversational loops and arbitrary verbosity, and in business settings it often necessitates constant supervision, additional orchestration, and manual enforcement of goals and constraints.

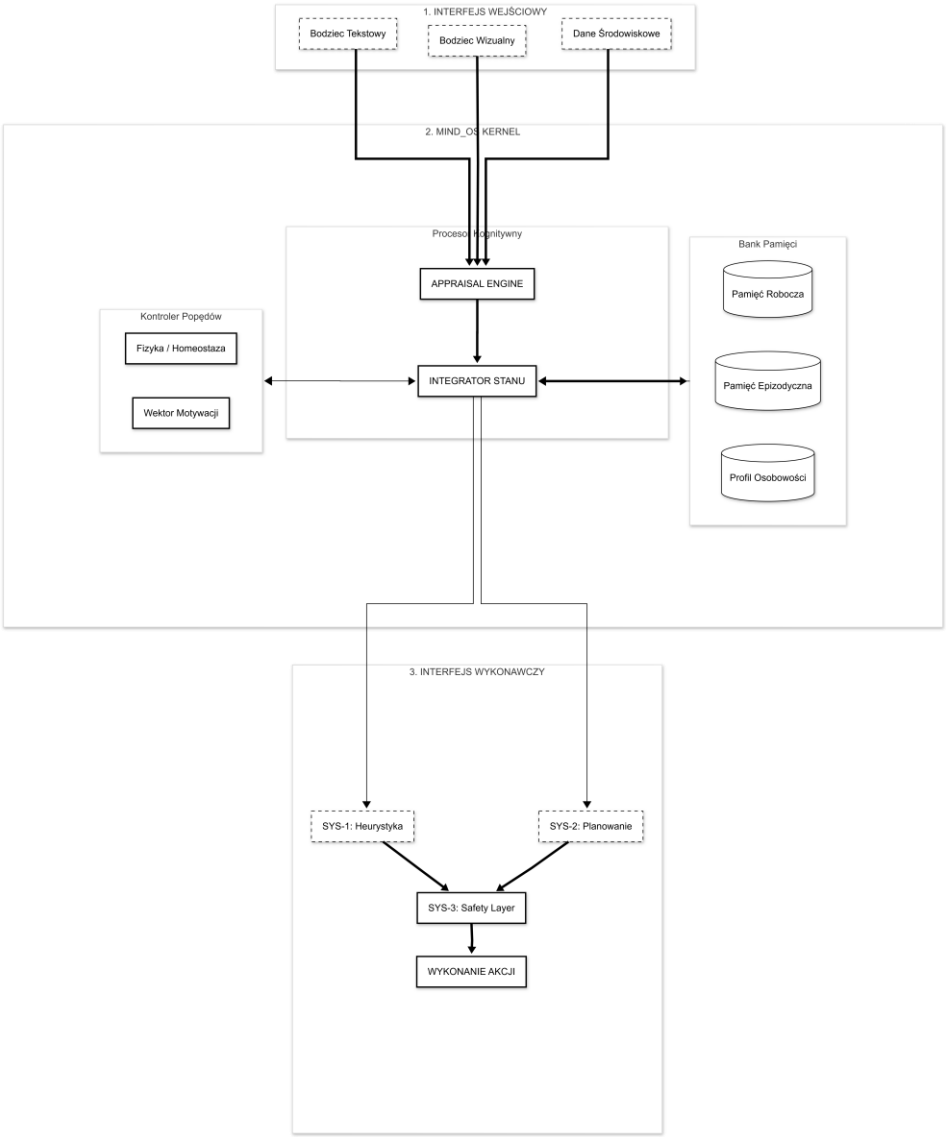
The MIND_OS architecture proposes to close this gap by implementing digital homeostasis—a mechanism in which the internal state is not merely a descriptive aid, but a real decision variable regulated over time. In this view, the system ceases to be only a text processor and becomes an instance that keeps its own parameters within an optimal range (e.g., energy, attention, affect/load), equipped with drives and memory that influence action selection. External stimuli are interpreted not only as data to be processed and verbalized, but also as factors that disrupt or stabilize the system's equilibrium—consuming energy, raising stress levels, demanding prioritization, or conversely enabling recovery and consolidation. In this way, autonomy is not merely "simulated" through prompting or an external control loop; it emerges from the necessity of state regulation: the system acts proactively to compensate disturbances, manage resources, and keep parameters within a desired range over time. This shifts the emphasis from one-off "answering" to continuous "functioning," fundamentally changing the quality and credibility of interaction: behavior becomes more consistent, less arbitrary, more grounded in priorities, and more resistant to drift resulting from the purely reactive nature of text generation.

3. Design assumptions and glossary

3.1. Assumptions

The MIND_OS architecture is built on a set of principles derived from observations in biology, cognitive psychology, and autonomous systems engineering, but deliberately simplified into a form that can be implemented and tuned in practical AI systems. The key objective is to create an instance that does not merely react to stimuli, but maintains continuity of state, manages resources rationally, and is capable of operating in a safe and predictable manner over a longer time horizon.

First, the system separates the "brain" from the "body." The "brain" denotes a lightweight, continuously operating mechanism responsible for maintaining internal state, updating memory, and making decisions at the level of control and strategy. It is designed to run continuously and at a relatively low computational cost. The "body," in turn, encompasses costly external actions such as acquiring information from the environment, interacting with services, executing tools, or performing operations that require more intensive resources (time, network, computation). The "body" is activated only when the decision made by the "brain" indicates a real need for action, which limits costs and strengthens control over operational risk.



Rys. 1. Block diagram of the MIND_OS architecture. An illustration of the information flow from perception, through the psychophysical and motivational layers, to the multi-level decision-making system.

Second, the internal state is persistent, and its most important components are stored locally. This means that the instance maintains continuity across sessions: it possesses a history of states, a trace of decisions, and episodic memory, rather than relying solely on the transient context of a conversation. Such persistence makes it possible to develop long-term tendencies, stabilize behavior over time, and build an operational identity (e.g., preferences, sensitivities, response patterns), which is unattainable for purely "prompt-based" systems.

Third, decisions carry costs and are governed by a principle of economy. The instance does not assume that every situation should be analyzed "as intelligently as possible," because under conditions of continuous

stimulus inflow this would lead to inefficiency and instability. Instead, decisions depend on drives, the current state (energy, attention, mood), and costs (time, risk, cognitive load). The system deliberately optimizes its behavior so that in typical situations it selects solutions that are sufficiently good and inexpensive, and only in critical situations activates more costly analytical mechanisms.

At this point, the architecture refers to the dual-process theory of Daniel Kahneman. By default, a fast, reactive, and economical mode operates, based on heuristics and decision thresholds, which enables action to be taken at minimal resource cost. When the significance of a stimulus increases (e.g., the emergence of risk, high uncertainty, goal conflict, or potentially large consequences), the system escalates to an analytical mode: slower, more deliberative, and more energy-expensive, but capable of more precise comparison of options and prediction of outcomes. This is a deliberately implementable interpretation of Kahneman's model—not an attempt to faithfully reproduce human psychology, but a practical mechanism for controlling decision costs and allocating computational resources.

Fourth, stimulus processing is staged and measurable in nature. Instead of a single, opaque "input → response" step, the instance passes through successive phases: stimulus perception, affective and cognitive appraisal, integration of the result into the internal state, selection of the decision mode, execution of action, and memory recording. This separation makes it possible to tune the parameters of each stage independently, diagnose errors, control sources of risk, and better understand why the system arrived at a given decision.

Fifth, there exists an independent safety control mechanism that is not subject to momentary fluctuations in mood or arousal. This means that even under high arousal, strong affect, or an increase in specific drives, the instance remains constrained by a set of hard rules and policies. Safety is treated as a supervisory layer: it filters intentions, limits actions, and blocks undesirable behaviors before they are translated into real operations in the external environment. As a result, the system maintains coherence and predictability of behavior, while operational risk is reduced independently of the dynamics of the internal state.

As a result, MIND_OS constitutes an attempt to build a practical architecture of a "digital instance" with a continuous, persistent state, which makes decisions in an economical, staged, and controllable manner, while maintaining clear safety boundaries and the ability to be extended with additional functional layers.

3.2. Core concepts

Homeostasis – the tendency to keep parameters within defined ranges. In a digital system, homeostasis is not "biology," but a control rule: deviations from the optimum generate pressure on decisions.

Drives – numerical motivational variables (e.g., curiosity, safety, affiliation, dominance, libido, energy). Drive values influence what the instance considers urgent and worthwhile.

Mood (PAD) – a vector of affective state described along three dimensions: pleasure (valence), arousal, and sense of control (dominance). The PAD model allows emotionality to be represented in a continuous rather than discrete manner.

Temperament – relatively stable traits that modulate reactions to stimuli, such as sensitivity, reactivity, sociability, or analytical inclination. Temperament does not change from minute to minute; it constitutes a slower "personality" layer.

Episodic memory – a record of events of elevated intensity along with metadata (time, context, emotional trace, significance). Episodic memory is the foundation of continuity and learning from experience.

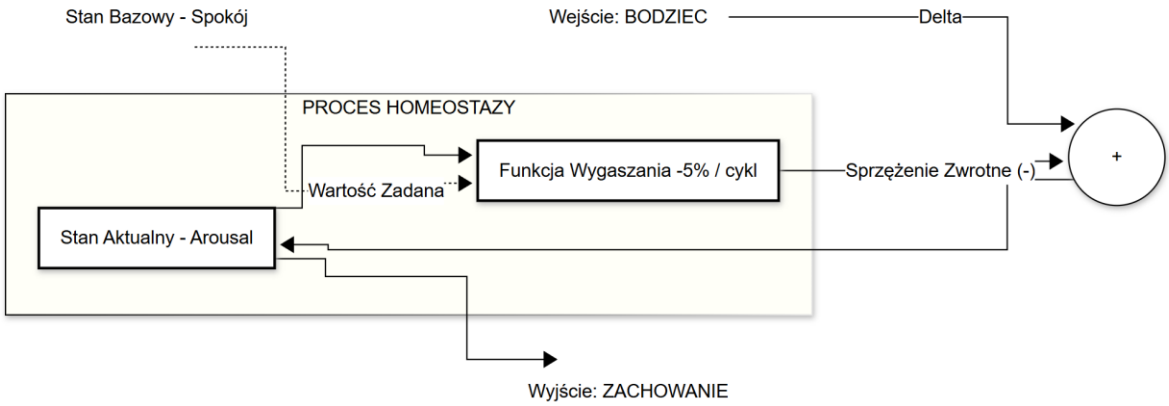
Narrative – the process of consolidating episodes into a coherent account and updating the self-model. Narrative plays a role analogous to "sleep": it organizes experiences, assigns them meaning, and modifies long-term priorities.

4. Psychophysical model: state, inertia, and costs

4.1. Emotional inertia and return to baseline

In MIND_OS, moods are not a switch but a dynamic state. Each stimulus generates a delta in the PAD vector, after which a decay mechanism is triggered, gradually returning the parameters to their baseline values. This approach prevents the instance from becoming "frozen" in extreme states and allows for credible variability.

Inertia also serves a function in stabilizing communication. The recipient of the interaction can see that the instance has emotional continuity: if a series of stressors occurs, the state does not immediately return to neutrality, but instead fades out gradually. From a control perspective, this functions as a low-pass filter—it limits abrupt jumps and makes behavior easier to predict.



Rys. 2. Block diagram of the homeostatic regulation loop. An illustration of the negative feedback loop responsible for the gradual reduction of arousal (decay) and stabilization of the internal state relative to the baseline value.

4.2. Cognitive metabolism and resource economy

Every stimulus processing carries a cost. MIND_OS assumes that resources (energy and attention) are limited, and that their level influences the style of operation. When resources are high, the instance can afford exploration, reflection, and more complex reasoning. When resources decline, priority shifts toward regeneration and risk minimization.

Cognitive metabolism is realized through differentiation of costs across decision modes. The fast (heuristic) mode is inexpensive and used by default, whereas the deep (reflective) mode is activated selectively—for example when a stimulus is important, conflicting, risky, or when the consequences of error are high.

This solution resembles human cognitive economy: a person does not analyze everything formally, but chooses when it is "worth thinking slowly."

4.3. Biological priorities and hard rules

In the architecture, priorities similar to biological ones are adopted. If the safety parameter falls below a threshold, the instance automatically restricts behaviors related to reward seeking (e.g., pleasure, flirtation, risky exploration). This is not "morality," but a simple control mechanism: survival takes precedence over reward.

The second group consists of hard safety rules. Regardless of mood, the instance does not undertake actions that conflict with the adopted principles. In practice, this means a layer that filters intentions and responses—analogous to how industrial systems apply safety interlocks independent of the control logic.

5. Perception and environmental context

5.1. The instance life loop

MIND_OS assumes that the instance is not active solely on demand. In the background, a life loop operates, cyclically collecting signals from the environment, updating the state, and—if conditions are met—initiating actions. Such a loop enables the existence of proactive behaviors: the instance may, for example, notice a significant change in context (new events, time of day, deteriorating conditions), assess its significance, and only then decide on a response.

The life loop is also the natural locus of homeostasis: even in the absence of new external stimuli, the internal state slowly drifts toward baseline values, while resources regenerate or are depleted in accordance with the parameters of the metabolism.

5.2. Context sources as "senses"

Environmental context is represented as a data structure containing, among other elements, local time, time of day, weather conditions, and informational signals about ongoing events. It serves as a set of

simplified “senses”: the instance receives a regular stream of data about the world, which can influence mood, motivation, and priorities.

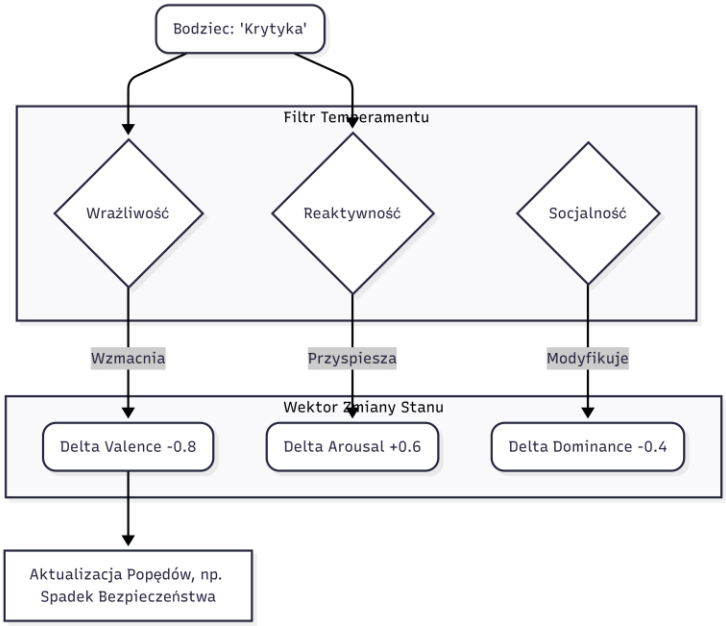
Importantly, context is not treated as a decorative addition to the prompt. In MIND_OS, context is a control element: it changes the probability of certain decisions. For example, late hours may increase fatigue, deteriorating weather may reduce pleasure, and a sudden increase in threats in the news may raise the safety drive.

5.3. Stimulus selection and noise reduction

The informational environment is noisy, therefore the architecture assumes stimulus filtering. The instance does not attempt to “read everything.” Instead, it selects a small number of signals with high novelty, high relevance, or alignment with current drives.

The selection mechanism serves a dual role. First, it protects cognitive resources from overload. Second, it creates a characteristic perceptual style: an instance with high curiosity will more often select new and unfamiliar stimuli, whereas an instance with a high safety drive will prefer warning and stabilizing signals.

6. Appraisal: from stimulus to state change



Rys. 3. Cognitive appraisal process (Appraisal). Transformation of an objective stimulus into a subjective change of the internal state, mediated by temperament parameters.

6.1. Affective appraisal as a function of meaning

After a stimulus is collected, the affective appraisal process is triggered. Its role is to answer the question: “what does this mean for me?”. A key feature of appraisal is that it does not describe the world “objectively,” but interprets it from the perspective of the instance, its goals, and its needs.

The result of appraisal is a delta in the PAD model along with auxiliary indicators (e.g., intensity, confidence of assessment). The delta is then integrated into the psychophysical state, taking into account inertia and current resources.

6.2. The role of temperament and response modulation

Temperament differentiates reactions to the same stimuli. High sensitivity increases the amplitude of mood change, while high reactivity shortens the rise time of the response. Sociability may amplify the positive impact of affiliative stimuli, whereas analytical inclination increases the tendency to activate the reflective mode.

From an implementation perspective, temperament functions as a multiplier and a filter: it does not generate content, but modifies the parameters of dynamics. This makes it possible to maintain a coherent “character” of the instance without the need to manually program dozens of exceptions.

6.3. Coupling with drives and homeostasis

Affective appraisal influences not only mood, but also drives. Stressful stimuli raise the safety drive and reduce the tendency toward exploration; rewarding stimuli may increase affiliation or curiosity. In this way, appraisal functions as a mechanism of motivational control rather than merely a description of emotions.

Feedback is crucial for autonomy. Drives influence which stimuli are selected (selection), and stimuli influence drives (updating). This creates dynamics similar to biological ones: the instance perceives the world in a way conditioned by its needs, while at the same time the world modifies those needs.

7. Drives as the engine of motivation

7.1. Why drives are necessary

Without drives, the system has no internal pressure to act. In MIND_OS, drives serve as internal control variables that determine what is urgent, what is worthwhile, and which behaviors are currently preferred.

Unlike “goals” encoded in instructions, drives are dynamic. They change with experience, context, and resources. As a result, the instance can exhibit behaviors such as satisfying curiosity, seeking safety, striving for affiliation, or reducing arousal.

7.2. Change mechanics: stimuli, metabolism, time

Drives are updated through three channels. The first is stimuli: each perception and appraisal can produce a change in one or more drives. The second is metabolism: over time, some drives naturally decrease or increase (e.g., energy may regenerate, curiosity may grow under monotony). The third channel is action: executed actions have a cost and may yield a reward, which directly affects drive values.

Understood in this way, drives function as a mechanism of motivational homeostasis. The instance strives to keep values from remaining in extreme states, while simultaneously using temporary deviations as signals for action.

7.3. Drive priorities and conflicts

Realistic behavior requires conflicts. Curiosity may compete with safety, affiliation with dominance, and pleasure with the need for regeneration. MIND_OS resolves conflicts through a prioritization function: each drive contributes its vote to the assessment of an action's utility, and the resulting outcome depends on current values and thresholds.

In practice, this enables richer behavior: the same instance may be exploratory and talkative under favorable conditions, and restrained and focused on resource protection under stress.

8. Decision systems: heuristics, reflection, and meta-control

8.1. Fast system: reactions and heuristics

The first decision layer is the fast system, responsible for immediate reactions. It operates on heuristics: simple rules that link state and stimulus to a response. The fast system is computationally inexpensive and performs well in situations where the cost of error is low or where time is critical.

In MIND_OS, the fast system draws, among other things, on dominant drives, the current mood, and simple stimulus classifications. This allows the instance to respond efficiently to the majority of interactions without activating costly reflection.

8.2. Reflective system: analysis and planning

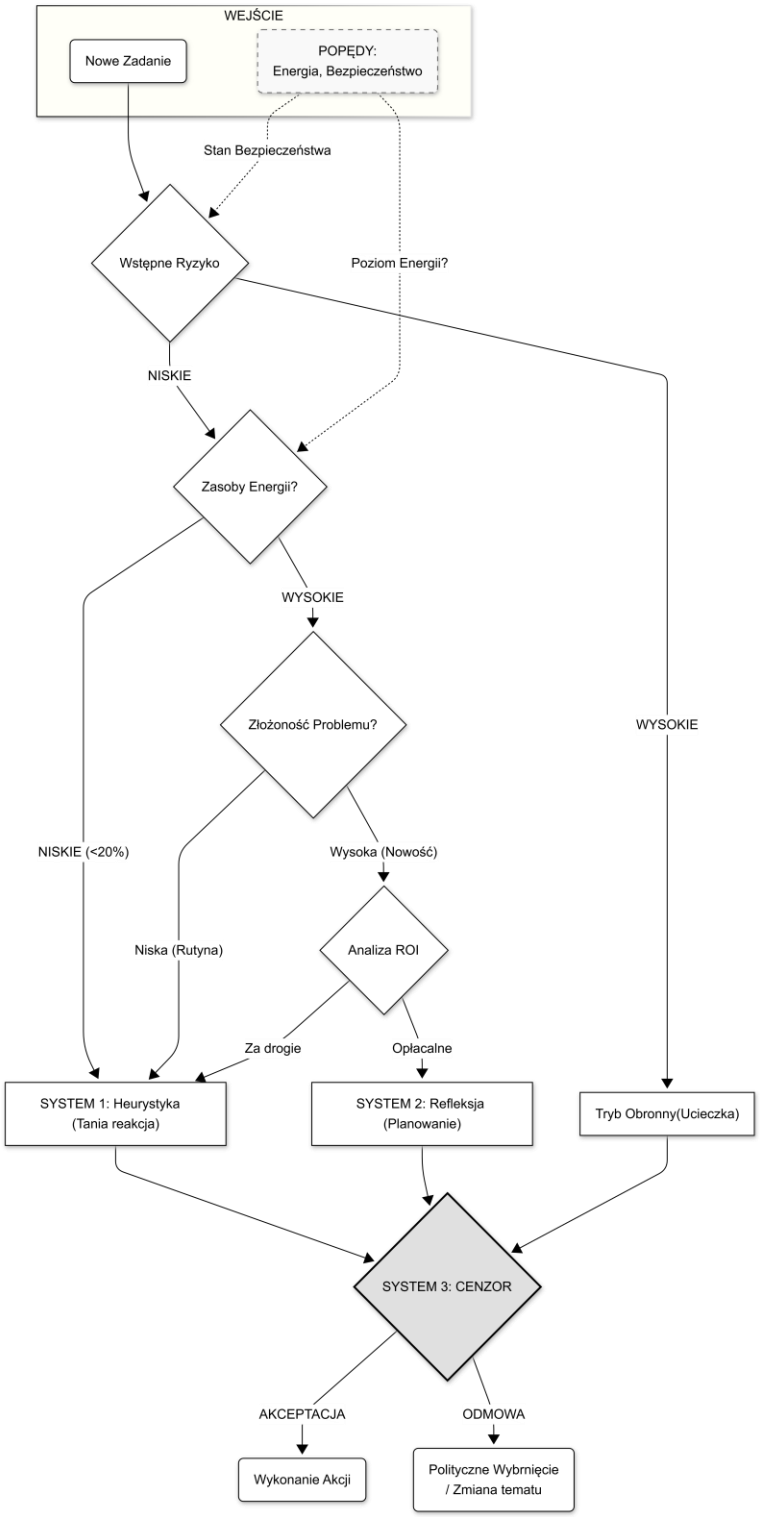
The second layer is the reflective system, activated selectively. Its purpose is to analyze contradictions, risks, and consequences. The reflective system can generate an action plan, verify hypotheses, and integrate a larger number of information sources.

The activation of reflection is an economic decision. The instance evaluates whether the benefit of higher decision quality outweighs the energetic and temporal cost. A high analytical inclination increases the frequency of reflection, while low resources constrain it.

8.3. Meta-control system: when to think slowly

The third layer serves a metacontrol function: it manages switching between the fast and the reflective systems. Metacontrol uses indicators such as stimulus significance, assessment uncertainty, risk level, conflict with memory, and current resources.

This design makes it possible to avoid two extremes: constant “overthinking” (where everything is analyzed exhaustively) and blind reactivity (where the instance never verifies its own heuristics). In practice, metacontrol is crucial for stable behavior in an informational environment.



Rys. 4. Metacontrol and Cognitive Metabolism Logic. The decision router algorithm that dynamically switches the instance between the reactive (inexpensive) mode and the reflective (costly) mode depending on available energy and task complexity.

9. Memory: episodes, knowledge, and continuity of identity

9.1. Working memory and short-term context

On a short time scale, the instance operates on working memory: the current conversation context (Message Window), recent stimuli, and the current state. Working memory is constrained by a token limit and is overwritten through a sliding window mechanism. This is desirable, as it protects the system from cognitive overload and ensures the fluency of interaction in the “here and now.”

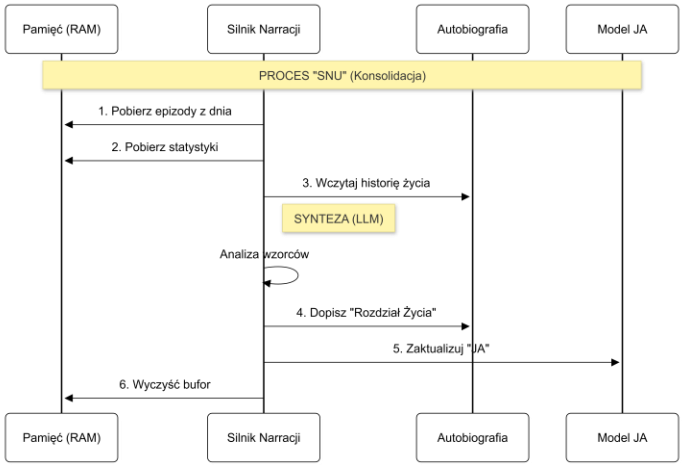
9.2. Episodic memory and emotional trace

Episodic memory is not a simple event log, but a vector-based repository of experiences. The system persistently stores only those events that exceed an intensity threshold (defined in the Appraisal engine)—that is, events that were important, surprising, or strongly emotionally charged. Each episode contains metadata of time and context, as well as an affective trace (the PAD state at the moment of storage). As a result, the instance not only “remembers” that something happened, but also “how it was experienced,” which enables associative recall based on emotional similarity rather than solely on keywords.

9.3. Knowledge memory and stable representations

In addition to episodes, the system stores structured semantic knowledge: facts, preferences, user profiles, and conclusions generated during the process of nightly consolidation (narrative). This layer changes more slowly and constitutes the foundation of identity. While working memory disappears when a session ends, knowledge memory builds a persistent world model of the instance, which evolves with each daily cycle.

10. Narrative and consolidation: The System Sleep



Rys. 5. Daily cycle and narrative mechanism (“Sleep”). The process of consolidating short-term memories into a persistent autobiographical structure and updating the identity model (Self-Model) based on experience.

10.1. Why narrative is needed in an AI system

Narrative is the mechanism that transforms raw experiences into a coherent self-model. Without narrative, the instance would possess episodes, but would be unable to generalize them or derive lasting conclusions. Narrative organizes the material: it groups events, selects key motifs, identifies patterns, and records them as elements of identity.

In practice, narrative functions as a daily cycle. At defined intervals, the instance creates a “chapter of the day,” in which it summarizes significant episodes, describes changes in state, and updates its self-description.

10.2. Updating the self-model and neuroplasticity

The self-model is a long-term structure: it contains stable traits, preferences, and beliefs about one’s own behavior. Narrative can adjust this model. If the instance consistently experiences similar situations (e.g., recurring stressors), it may increase caution; if it experiences rewards for exploration, it may become more curious.

This approach is a digital analogue of neuroplasticity: it is not about “training” the language model, but about learning an action policy through changes in control parameters (temperament, thresholds, drive weights).

11. Social layer and shared context

11.1. Social mode as a separate action channel

MIND_OS предусматривает a separate layer responsible for behavior in the social environment: group conversations, public channels, discussions, and narrative confrontations. This layer is not identical to language generation; its role is to decide when and how to speak, and when to remain in the background.

Crucial here is the management of reputation and risk. The instance evaluates not only the content of a stimulus, but also the potential social consequences of its expression.

11.2. Multi-instance architecture and the “council”

In multi-agent scenarios, instances can be combined into a cooperative structure. Each entity may possess its own temperament and drives, while simultaneously operating within a shared event space. Such an arrangement enables diversity of interpretation and allows simulation of a “council” dynamic, where agents discuss privately and then publish independent positions.

In practice, this solution enables scalable analysis: instead of a single monolith, a team of specialized entities emerges, capable of sharing tasks among themselves.

12. Monitoring, audit, and operational safety

12.1. Monitoring as an “autonomic system”

Continuous monitoring provides visibility into the state of the instance: mood levels, energy, dominant drives, frequency of reflection, and episode intensity. Monitoring plays the role of an autonomic system—it detects undesirable states (e.g., prolonged stress, excessively low resource levels) and enables corrective responses.

From an engineering perspective, monitoring is critical for debugging. In “living” systems, errors are rarely single events; more often they are the result of dynamics, which is why time series and historical traces are required.

12.2. Safety layer and behavioral compliance

Regardless of mood and drives, the instance must remain compliant with safety policies. This is achieved through multilayered control: intention filtering, blocking of selected actions, and constraints at the tooling layer. Such an arrangement resembles safety systems in automation—the control logic cannot bypass a safety interlock.

An important element is also auditing: recording decisions and their justifications (in the form of metadata), which makes it possible to analyze errors and improve thresholds, weights, and heuristics.

13. Information flow: from stimulus to action

The entire cycle can be summarized as follows: (1) perception collects stimuli and context, (2) appraisal interprets the significance of the stimulus for the instance and generates changes in affect, (3) the state integrator updates psychophysical parameters taking inertia and resources into account, (4) the motivational module updates drives and determines dominant needs, (5) metacontrol selects the decision system, (6) a decision about action or utterance is made, and (7) outcomes and consequences are recorded in memory, with high-intensity episodes sent to narrative consolidation.

This flow is deliberately modular. Each stage has its own parameters and can be tuned independently, which in practice accelerates development in an R&D mode.

14. Limitations, risks, and development directions

MIND_OS is a pre-alpha architecture and carries risks typical of dynamic systems. First, parameters may require tuning to avoid oscillations (e.g., excessive switching between decision systems) or “lock-ins” (maintaining an excessively high safety level). Second, stimulus selection must balance novelty and

relevance so as not to lead to an informational bubble. Third, with multiple instances, computational economy and synchronization become increasingly important.

A natural direction of development is the reinforcement of three areas: (1) a full homeostatic “tick” operating independently of stimuli, (2) formalization of the influence of narrative on long-term drive weights and temperament, and (3) extension of evaluation metrics to measure behavioral stability, adaptability, and safety over time.

15. Conclusion

MIND_OS represents an attempt to move from reactive agents toward autonomous digital entities that possess continuity, motivation, and action cost. The key innovation is the separation of the language generation layer from the control layer: behavior is determined by the psychophysical state, drives, memory, and metacontrol, while language serves as a tool of expression.

Such an architecture is particularly useful in informational environments where priorities, resistance to overload, and long-term coherence are critical. The pre-alpha version confirms the feasibility of the concept and provides a solid foundation for further research and development work.